SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1711

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour
QUESTIONS: 4 **Total: Marks:40**

Annexure A

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Code of Conduct:

I Kishore S.P (192511081) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning

Use supervised inductive learning to learn the relationship between patient features and diabetes status.

- Target: Diabetes (Yes/No)
- Key features:
  - Glucose: strong indicator of diabetes.
  - BMI: associated with insulin resistance.
  - Age: diabetes risk generally increases with age.
  - Blood pressure/family history: useful additional risk factors.

(b) Class Imbalance

Use SMOTE on the training set because diabetic cases are fewer. It creates synthetic minority examples and improves recall without discarding many non-diabetic records.

(c) 70:30 Split & Preprocessing

Use a 70% training / 30% testing stratified split. This provides enough data for training while keeping a sufficiently large test set.

Preprocessing:

- Encode Family History: Yes = 1, No = 0.
- Handle missing values using imputation.
- Standardise numerical features for Logistic Regression.
- Decision Trees do not require normalisation.
- Apply SMOTE only to training data.

Task 2: Decision Tree

(a) Tree Construction

Use a Decision Tree with Gini Index or Information Gain.

Example:

```
          Glucose
         /      \
       Low      High
      /           \
    BMI           Age
   /  \          /  \
 No   Risk     Risk  Yes
```

The feature with the highest Information Gain / lowest Gini impurity becomes the root. Typically, glucose is expected to be a strong root feature, but the actual dataset must determine the result.

(b) Evaluation

Use:

- Accuracy = correct predictions / total predictions
- Precision = TP / (TP + FP)
- Recall = TP / (TP + FN)
- F1-score = harmonic mean of precision and recall.

False Negative: diabetic patient predicted as non-diabetic.

To reduce false negatives:

- use class weights;
- apply SMOTE;
- lower the classification threshold where clinically appropriate;
- prioritise high recall/sensitivity.

This is important because missed diabetes cases can delay diagnosis and treatment.

(c) Post-Pruning

Use cost-complexity pruning:

$$R_\alpha(T)=R(T)+\alpha|T|$$

It removes unnecessary branches and reduces overfitting.
A pruned tree is generally preferable for clinical deployment if it maintains or improves validation performance while being simpler and more interpretable.

Task 3: Statistical Learning
(a) Logistic Regression
Use Logistic Regression as a statistical baseline.

$$P(Diabetes)=\frac{1}{1+e^{-z}}$$

Compare it with the Decision Tree using:

- Accuracy
- AUC-ROC

A statistical model is preferable when interpretability, probability estimates, and understanding predictor effects are important.
(b) Feature Importance
Possible top predictors:

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Age

If both models agree, this suggests these variables contain strong predictive information. However, feature importance does not prove causation.

Task 4: Reinforcement Learning
(a) MDP

$$MDP=(S,A,P,R,\gamma)$$

- States: Low risk, Moderate risk, High risk, Poor control, Improved.
- Actions: Diet, Exercise, Medication review, Monitor.
- Reward: Positive for health improvement and negative for deterioration/adverse outcomes.
- Transitions: Patient moves between health states according to treatment and health changes.

(b) Q-Learning
Update rule:

$$Q(s,a)\leftarrow Q(s,a)+\alpha[r+\gamma\max Q(s',a')-Q(s,a)]$$

Example with ($\alpha=0.5,\gamma=0.9$):
If:

$$Q(S_2,D)=0,\quad r=5$$

then:

$$
Q(S_2,D)=0+0.5(5)=2.5
$$

After another update where the next state's best Q-value is 2.5:

$$
Q(S_2,D)=2.5+0.5[5+0.9(2.5)-2.5] =4.875
$$

Another example:

$$
Q(S_3,Monitor)=0.5[2+0.9(4.875)] =3.194
$$

Train for at least 5 patient episodes.

Convergence criterion:

$$
\max |Q_{new}-Q_{old}|<0.001
$$

for several consecutive iterations.

(c) Final Policy

The policy is:

$$
\pi(s)=\arg\max_a Q(s,a)
$$

Example:

Patient State	Best Action
Low risk	Exercise
Moderate risk	Exercise
High risk	Diet
Poor control	Medication review
Improved	Monitor

Ethical Considerations

- Safety: RL must not prescribe medication autonomously.
- Bias: evaluate performance across patient groups.
- Explainability: provide reasons for recommendations.
- Human oversight: clinicians must approve treatment decisions.
- Privacy: protect patient medical data.
- Validation: test extensively before clinical deployment.

Conclusion

A suitable solution is Decision Tree + Logistic Regression for diabetes prediction, followed by a constrained Q-Learning framework for treatment recommendations. In clinical deployment, recall, safety, fairness and interpretability are more important than accuracy alone.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation